WEATHER PREDICTION USING MACHINE LEARNING

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Introduction

This Weather Prediction System leverages machine learning models to analyze real-time and historical weather data for accurate forecasting. It collects current weather conditions from the OpenWeatherMap API and processes historical data to train models for rainfall prediction and temperature/humidity forecasting. Using techniques like Recursive Feature Elimination (RFE) and Random Forest classifiers/regressors, the system identifies key weather parameters, improves accuracy, and generates short-term forecasts. The predictions, including rain probability, temperature variations, and humidity levels, are compiled into an intuitive report, providing users with a structured and insightful weather outlook

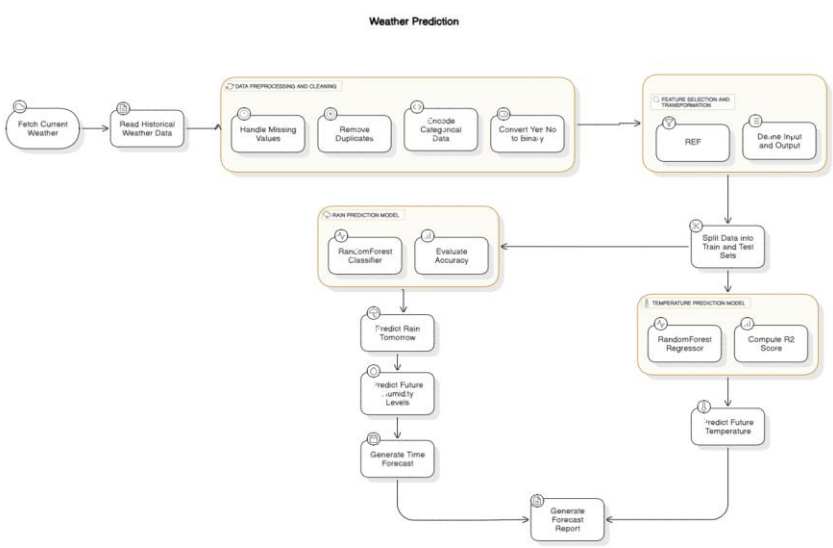
Scope of the project

The primary focus of this project is to develop a machine learning model that predicts weather conditions with improved precision. It incorporates real-time weather data retrieval using the OpenWeatherMap API, allowing users to access live temperature, humidity, wind speed, and pressure for any given location. Additionally, historical weather data is processed to identify patterns and trends that contribute to accurate forecasting. The system is designed to predict whether it will rain the next day using a classification model while also employing regression techniques to estimate future temperature and humidity levels. Feature selection techniques such as Recursive Feature Elimination (RFE) are implemented to enhance model performance, ensuring that only the most relevant weather attributes are considered. Furthermore, the project includes data visualization components, allowing users to interpret weather trends through graphical representations, confusion matrices, and accuracy comparisons.

Methodology

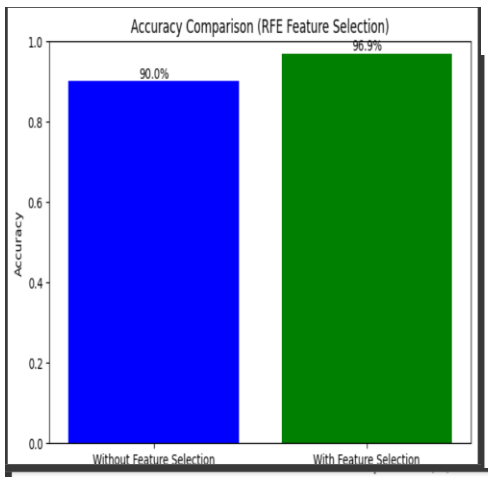
The project follows a structured approach, beginning with data collection and preprocessing, where missing values are handled, duplicate entries are removed, and categorical variables like wind direction are encoded numerically. Real-time weather data is integrated via API requests for accurate predictions. Feature selection using Recursive Feature Elimination (RFE) identifies the most relevant weather parameters, and the dataset is split into training and testing sets for model learning. A Random Forest Classifier predicts rainfall probability, while a Random Forest Regressor forecasts temperature and humidity. The trained model generates hourly and daily predictions, with wind direction converted into compass points for better interpretation. Model performance is evaluated using accuracy scores, confusion matrices, and feature importance visualizations to ensure reliable weather forecasting

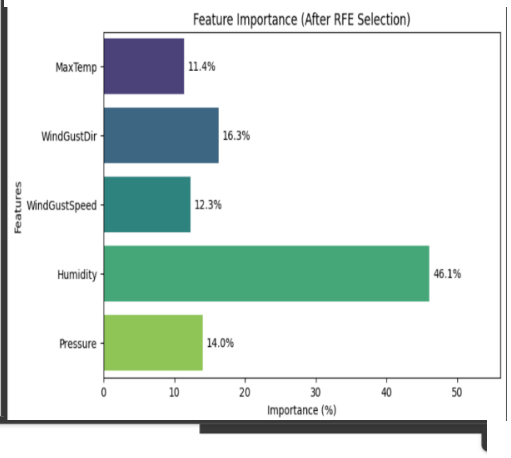
Architecture Diagram

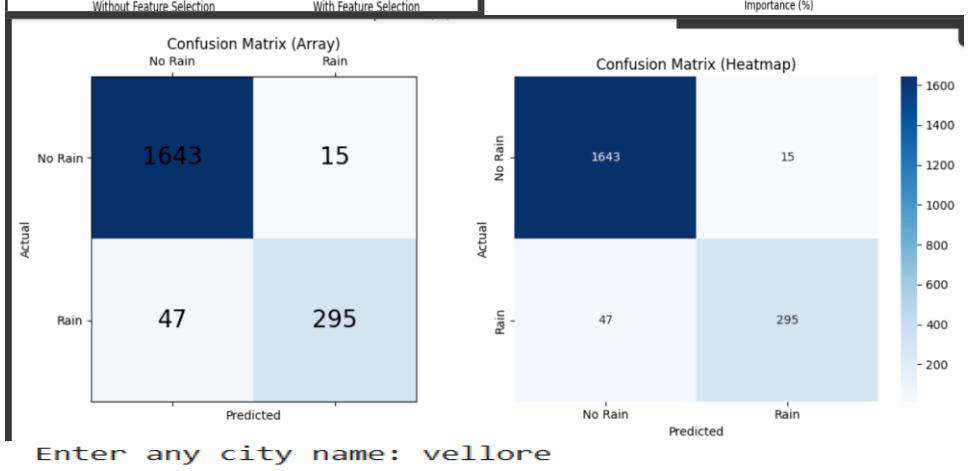


The weather prediction model begins by collecting real-time and historical weather data, which is then preprocessed by handling missing values, removing duplicates, and encoding categorical variables. Feature selection using Recursive Feature Elimination (RFE) identifies the most relevant parameters, and the dataset is split into training and testing sets. A Random Forest Classifier predicts rainfall probability, while a Random Forest Regressor forecasts temperature and humidity. Model performance is evaluated using accuracy scores and R^2 values. Based on these predictions, a time-based weather forecast is generated, and a final forecast report is created for easy interpretation

Result








```
Enter any city name: vellore

City: vellore, IN
Current Temperature: 37°C
Feels Like: 35°C
Humidity: 21%
Min Temperature: 37°C
Max Temperature: 37°C
Overall Model Accuracy: 0.97 (96.9%)
```

The system provides a comprehensive weather report, including the current temperature, humidity, wind speed, pressure, and an overall weather description based on real-time data. It predicts the possibility of rain using the trained classification model, helping users prepare in advance. Additionally, the regression models forecast temperature and humidity for the next five hours, giving users a short-term weather outlook. The output also includes detailed model performance metrics, confusion matrices, and feature importance visualizations, ensuring transparency in predictions

Conclusion

This project demonstrates the effectiveness of Machine Learning in weather forecasting by integrating real-time data with predictive modeling. By utilizing feature selection techniques, the accuracy of predictions is significantly improved, making the model both efficient and interpretable. The combination of classification and regression models ensures comprehensive weather analysis, providing users with reliable forecasts. As a future enhancement, deep learning techniques and real-time deployment can further improve the precision and scalability of the system, making it a valuable tool for various industries and individual users alike

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Acknowledgments/ References

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